Example 12.19

Using the equivalence property, show that $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$.

Solution

It can be obtained by using examples 12.15 and 12.16 that

$$p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p) \quad \text{... (1)}$$

$$\equiv (\neg p \vee q) \wedge (p \vee \neg q) \text{ (by Commutative Law)} \quad \text{... (2)}$$

$$\equiv (\neg p \wedge (p \vee \neg q)) \vee (q \wedge (p \vee \neg q)) \text{ (by Distributive Law)}$$

$$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \vee (q \wedge p) \vee (q \wedge \neg q) \text{ (by Distributive Law)}$$

$$\equiv \mathbb{F} \vee (\neg p \wedge \neg q) \vee (q \wedge p) \vee \mathbb{F}; \text{ (by Complement Law)}$$

$$\equiv (\neg p \wedge \neg q) \vee (q \wedge p); \text{ (by Identity Law)}$$

$$\equiv (p \wedge q) \vee (\neg p \wedge \neg q); \text{ (by Commutative Law)}$$

Finally (1) becomes $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$.

EXERCISE 12.2

1. Let p : Jupiter is a planet and q : India is an island be any two simple statements. Give verbal sentence describing each of the following statements.
   (i) $\neg p$ (ii) $p \wedge \neg q$ (iii) $\neg p \vee q$ (iv) $p \rightarrow \neg q$ (v) $p \leftrightarrow q$
2. Write each of the following sentences in symbolic form using statement variables p and q.
   (i) 19 is not a prime number and all the angles of a triangle are equal.
   (ii) 19 is a prime number or all the angles of a triangle are not equal
   (iii) 19 is a prime number and all the angles of a triangle are equal
   (iv) 19 is not a prime number
3. Determine the truth value of each of the following statements
   (i) If $6+2=5$, then the milk is white.
   (ii) China is in Europe or $\sqrt{3}$ is an integer
   (iii) It is not true that $5+5=9$ or Earth is a planet
   (iv) 11 is a prime number and all the sides of a rectangle are equal
4. Which one of the following sentences is a proposition?
   (i) $4+7=12$ (ii) What are you doing? (iii) $3^n \leq 81, n \in \mathbb{N}$
   (iv) Peacock is our national bird (v) How tall this mountain is!
5. Write the converse, inverse, and contrapositive of each of the following implication.
   (i) If x and y are numbers such that $x = y$, then $x^2 = y^2$
   (ii) If a quadrilateral is a square then it is a rectangle
6. Construct the truth table for the following statements.
   (i) $\neg p \wedge \neg q$ (ii) $\neg(p \wedge \neg q)$ (iii) $(p \vee q) \vee \neg q$ (iv) $(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$

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